

Smart Financial Services Kiosk with AI Advisory System

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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ABSTRACT

Smart Financial Services Kiosk with AI Advisory System is a web-based financial management system developed using Python, Flask, SQLite, HTML, CSS, and JavaScript that helps users manage their personal finances by providing secure login, financial tracking, AI-based financial advice, fraud detection, document management, and an interactive financial chatbot. The system allows users to register, securely log in, and monitor their income, savings, expenses, and financial goals through a professional dashboard. It also includes an Admin Dashboard for managing users, viewing activity logs, monitoring statistics, searching user records, generating financial charts, and exporting reports in PDF and Excel formats. The application emphasizes data security, ease of use, and efficient financial planning while helping users identify potential fraud attempts and organize important financial documents. With its modern interface and comprehensive features, the system provides a practical solution for personal financial management and digital banking assistance.

Keywords: Smart Financial Services Kiosk, AI Financial Advisor, Raspberry Pi 4 Model B, KYC, Document Upload, Fraud Detection, User History, Web-Based Interface.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

1. SYMBOLS:

Symbol	Meaning
V	Voltage
GB	Gigabyte
GHz	Gigahertz
API	Application Programming Interface
₹	Indian Rupee

2. ABBREVIATIONS:

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
SQL	Structured Query Language
UI	User Interface
PDF	Portable Document Format
RAM	Random Access Memory
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
OCR	Optical Character Recognition
OS	Operating System
Wi-Fi	Wireless Fidelity

3.NOMENCLATURE:

Term	Definition
Smart Financial Advisor and Secure Banking Assistant	A secure web-based financial management application that integrates AI-powered financial services.
AI Financial Advisor	Provides personalized financial recommendations based on user financial information.
Financial Chatbot	AI-powered chatbot that answers finance-related queries.
Fraud Detection	Module that analyzes suspicious financial messages and transactions to identify fraud.
Document Management	Module used to upload, organize, and securely store financial documents.
User Authentication	Secure login and registration system for authorized access.
Admin Dashboard	Administrative interface for monitoring users, activities, reports, and analytics.
Activity Log	Stores records of user activities performed within the application.
Flask	Python web framework used to develop the application.
SQLite	Lightweight database used to store user and financial information.
Groq API	AI service used to generate intelligent financial recommendations.

CHAPTER 1

INTRODUCTION

In today's digital era, technology has significantly transformed the way people manage their personal finances. Online banking, digital payments, investment platforms, and mobile financial applications have made financial transactions faster, more convenient, and easily accessible. However, the increasing dependence on digital financial services has also introduced several challenges, including financial fraud, cyber threats, poor financial planning, and the difficulty of managing financial records securely. Many individuals lack proper financial knowledge and struggle to make informed decisions regarding savings, budgeting, investments, and long-term financial goals.

To address these challenges, an intelligent and secure financial management system is essential. The Smart Financial Advisor and Secure Banking Assistant is a web-based application designed to provide users with a comprehensive platform for managing their personal finances while ensuring security and ease of use. The application integrates financial planning, AI-based financial advice, fraud detection, document management, and an interactive financial chatbot into a single system. It enables users to monitor their income, savings, expenses, and financial goals through an intuitive dashboard, helping them make better financial decisions.

The system is developed using Python, Flask, SQLite, HTML, CSS, and JavaScript, providing a lightweight, secure, and efficient architecture. It also includes a professional administrative dashboard that allows administrators to manage user accounts, monitor system activities, generate financial statistics, search user records, and export reports in PDF and Excel formats.

The primary aim of this project is to promote financial awareness, improve personal financial management, and enhance digital banking security through intelligent technologies. By combining financial advisory services with fraud prevention and secure document management, the system offers a reliable and

user-friendly solution for individuals seeking effective financial planning and secure digital banking assistance.

The proposed project focuses on developing a functional prototype that demonstrates the integration of artificial intelligence, digital document handling, fraud risk evaluation, and user activity management in a single financial services kiosk. The system provides a compact for demonstrating an interactive approach to basic digital financial services.

1.1 PROJECT BACKGROUND AND MOTIVATION

The rapid development of digital technology has changed the way financial services are accessed and managed. Web-based platforms are increasingly used for financial guidance, document submission, and other financial service processes. However, accessing different services through separate platforms can make the overall process less convenient for users. This creates a need for an integrated system that brings essential financial service functions together in a single interface.

The Smart Financial Services Kiosk with AI Advisory System is developed to provide an interactive platform that integrates basic financial guidance and related digital financial service processes. The system combines an AI Financial Advisor, KYC and Document Upload, Fraud Detection, and User History modules through a common web-based interface. This allows users to access the available services in an organized and convenient manner.

The main motivation behind this project is to demonstrate how artificial intelligence and digital technologies can be integrated into a compact kiosk-based platform. The system is designed to operate using a Raspberry Pi 4 Model B with 4 GB RAM as the main computing platform. By integrating multiple functional modules into a single system, the project aims to develop a functional and integrated prototype of an interactive financial services kiosk.

1.2 PROBLEM STATEMENT

The increasing use of digital financial services has created a need for systems

that allow users to access basic financial guidance and related services through a simple and integrated platform. In many cases, financial guidance, KYC document submission, fraud checking, and user activity records are handled through separate processes or platforms, making it less convenient for users to access these functions together.

Therefore, there is a need to develop a single interactive system that integrates these essential functions into one platform. The proposed Smart Financial Services Kiosk with AI Advisory System addresses this requirement by combining an AI Financial Advisor, KYC and Document Upload, Fraud Detection, and User History modules through a common web-based interface. The system is designed to operate using a Raspberry Pi 4 Model B with 4 GB RAM, providing a compact platform for demonstrating integrated digital financial services.

1.3 PROJECT OBJECTIVES

The primary objective of the Smart Financial Advisor and Secure Banking Assistant is to develop a secure, intelligent, and user-friendly web application that helps users manage their personal finances effectively while improving awareness of digital banking security. The system aims to integrate financial planning, fraud prevention, and administrative monitoring into a single platform.

- The specific objectives of the project are:
- To develop a secure web-based financial management system using Python, Flask, SQLite, HTML, CSS, and JavaScript.
- To provide secure user registration and authentication for protecting user accounts and financial data.
- To enable users to record and manage their income, savings, expenses, and financial goals through an interactive dashboard.
- To provide AI-based financial advice that helps users make informed financial decisions based on their financial information.
- To implement a financial chatbot that answers common finance-related

queries and offers basic financial guidance.

- To detect potentially fraudulent messages and suspicious financial activities, thereby increasing user awareness of online scams.
- To provide secure document management for uploading, storing, and organizing important financial documents.
- To develop a professional administrator dashboard for managing users, monitoring activities, and viewing system statistics.
- To generate graphical reports and enable data export in PDF and Excel formats for efficient reporting and analysis.
- To promote financial literacy, encourage responsible financial planning, and improve the overall user experience through a secure and modern web application.

1.4 METHODOLOGY

The development of the Smart Financial Advisor and Secure Banking Assistant follows a systematic software development methodology to ensure that the application is secure, reliable, and user-friendly. The project is implemented through the following phases:

Step 1: Requirement Analysis

In this phase, the system requirements are gathered by identifying the needs of both users and administrators. Functional requirements such as user registration, login, financial dashboard, AI financial advisor, chatbot, fraud detection, document management, and report generation are defined. Non-functional requirements, including security, performance, usability, and reliability, are also considered.

Step 2: System Design

The system architecture, database schema, and user interface are designed. The application is divided into different modules such as user authentication, financial management, AI advisor, fraud detection, document management, and the admin dashboard. The database is designed using SQLite to store user

information, financial records, activity logs, and uploaded documents.

Step 3: Implementation

The application is developed using Python with the Flask framework for the backend. SQLite is used as the database, while HTML, CSS, and JavaScript are used to create a responsive and user-friendly interface. The implementation includes secure login, user registration, financial dashboards, chatbot integration, fraud detection, document upload, data visualization, and report generation in PDF and Excel formats.

Step 4: Testing

The developed system is thoroughly tested to ensure that all modules function correctly. Unit testing, integration testing, and system testing are performed to identify and fix errors. Testing also verifies authentication, data validation, report generation, fraud detection, and overall system performance.

Step 5: Deployment

After successful testing, the application is deployed on the target system. The database is configured, required software dependencies are installed, and the application is made accessible through a web browser. This phase ensures that users can securely access all system features.

Step 6: Maintenance

After deployment, the system is continuously monitored and maintained. Bugs are fixed, security updates are applied, and new features can be added based on user feedback and future requirements. Regular maintenance ensures that the application remains secure, efficient, and compatible with evolving technologies.

Component Technology	Used
Programming Language	Python
Web Framework	Flask
Database	SQLite
Frontend	HTML, CSS, JavaScript

IDE	Visual Studio Code
Operating System	Raspberry Pi OS / Linux
Report Generation	PDF and Excel
Version Control	Git (optional)

Table 1.1 Development Tools Used

This methodology ensures a structured development process, resulting in a secure, efficient, and scalable financial management application that meets the project's objectives.

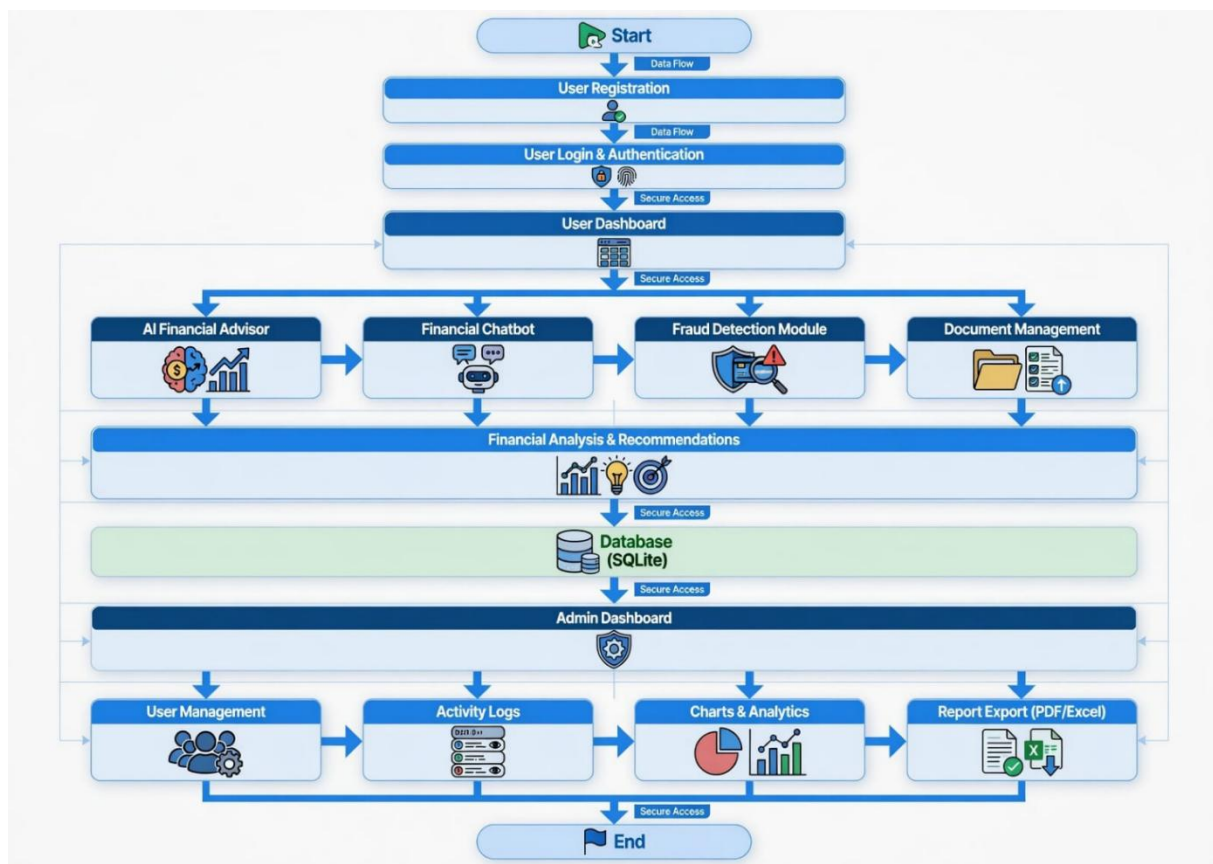


Fig 1.1 Working Methodology

CHAPTER 2

LITERATURE REVIEW

1. Innovative Financial Technologies: Strengthening Compliance, Secure Transactions, and Intelligent Advisory Systems Through AI-Driven Automation and Scalable Data Architectures [1]

The authors studied the use of artificial intelligence and automated technologies in modern financial systems. The paper highlights the role of AI-driven automation in improving financial advisory services, transaction security, and system efficiency. The study shows that intelligent technologies can support faster decision-making and more personalized financial services. However, secure data management and system scalability remain important challenges.

2. A Multi-Layered AI-IoT Framework for Adaptive Financial Services [2]

The authors proposed a multi-layered framework that integrates artificial intelligence and IoT technologies for adaptive financial services. The framework consists of input, intelligence, and experience layers to process user information and provide intelligent financial services. The study demonstrates how AI-based analysis can support personalized services, automated decision-making, and real-time risk management. However, the integration of multiple technologies increases system complexity.

3. The Response of Islamic Financial Service to the COVID-19 Pandemic: The Open Social Innovation of the Financial System [3]

The authors examined how digital financial technologies supported financial services during the COVID-19 pandemic. The study highlights the importance of digital platforms in improving financial accessibility and maintaining financial services when direct physical interaction is limited. The paper concludes that technology-based financial systems can improve service accessibility and support the development of innovative financial service models.

4. Kiosk Banking: The Challenges to and Implications of Building an Inclusive Financial Infrastructure [4]

The authors studied kiosk banking as an approach for providing financial services through accessible self-service infrastructure. The study highlights the potential of kiosk-based systems to improve access to financial services and reduce dependence on conventional service methods. It also discusses challenges related to system operation, user accessibility, infrastructure, and service reliability. The study shows that kiosk systems can support inclusive financial service delivery when properly implemented.

5. Intelligent Finance: The Evolution and Impact of AI-Driven Advisory Services in FinTech [5]

The authors explored the development of AI-driven financial advisory services in the FinTech sector. The study explains how artificial intelligence can analyze user information and provide automated financial guidance. AI-based advisory systems can improve accessibility, efficiency, and personalization compared to traditional financial advisory methods. However, the reliability of recommendations and responsible handling of user data remain important considerations.

6. AI-Powered Wealth Management: Transforming Financial Literacy, Personalized Investments, and Risk Assessment Through Robo-Advisors and Predictive Analytics for the Future of Finance [6]

The authors examined the use of AI in wealth management, financial literacy, personalized investment guidance, and risk assessment. The study highlights the role of robo-advisors and predictive analytics in providing personalized financial recommendations based on user information. The results indicate that AI-based systems can improve access to financial guidance and support informed financial decision-making. However, the quality of the recommendations depends on the accuracy of user data and the reliability of the analytical system.

7. Financial Advisory and Management System [7]

The authors developed a financial advisory and management system to assist users in managing financial information and obtaining financial guidance. The system focuses on organizing user financial data and providing advisory support through a digital platform. The study demonstrates the usefulness of integrating financial information management and advisory functions within a single system. This approach improves convenience by allowing users to access related financial functions through one platform.

8. Leveraging IoT-Driven Customer Intelligence for Adaptive Financial Services [8]

The authors studied the use of IoT-based customer information for developing adaptive and personalized financial services. The system collects information from different customer interaction points and uses the data to improve service personalization. The study highlights the potential of real-time customer intelligence in developing responsive financial systems. However, data privacy and the integration of different data sources remain major challenges.

9. Proceedings of the International Conference on Emerging Trends in Business & Management (ICETBM 2023) [9]

The conference proceedings presented studies related to emerging technologies and their applications in modern business and management systems. The works highlight the growing importance of digital platforms, artificial intelligence, automation, and data-based decision-making. These technologies can improve service efficiency and user interaction across different sectors, including financial services. The studies demonstrate the increasing role of intelligent digital systems in modern service delivery.

This is in the **same short literature-review style** as your previous project report: title → what the authors studied → method/focus → result → limitation where relevant. I have kept it to the **9 unique references** shown in your sheet; the repeated “Innovative Financial Technologies” entry should not be counted .

CHAPTER 3

SYSTEM ARCHITECTURE AND IMPLEMENTATION

3.1 OVERALL SYSTEM ARCHITECTURE

The Smart Financial Advisor and Secure Banking Assistant is developed as a web-based application that integrates user authentication, financial management, AI-based financial advisory, financial chatbot, fraud detection, document management, and administrative monitoring into a single platform. The system architecture is designed to provide users with secure access to all financial services through a unified web interface.

The application is built using the Flask web framework with Python as the backend programming language. SQLite is used as the database to store user information, financial records, uploaded documents, and activity logs. The frontend is developed using HTML, CSS, and JavaScript, providing a responsive and user-friendly interface that allows users and administrators to access the system through a web browser.

Users interact with the application through the login and registration pages. After successful authentication, they are redirected to a personalized dashboard where they can view and manage their financial information, including income, savings, expenses, and financial goals. The dashboard also provides access to the AI Financial Advisor, Financial Chatbot, Fraud Detection module, and Document Management module.

The AI Financial Advisor analyzes the user's financial information and provides personalized financial recommendations to support better financial planning and decision-making. The Financial Chatbot answers finance-related queries and offers general financial guidance. The Fraud Detection module analyzes user-submitted messages or transaction details to identify potential fraud risks and alert users to suspicious activities. The Document

Management module enables users to securely upload and organize important financial documents.

The system also includes a comprehensive Admin Dashboard that allows administrators to monitor registered users, manage user information, view activity logs, analyze financial statistics, generate graphical reports, and export reports in PDF and Excel formats. These administrative features support efficient system monitoring and data management.

The application is developed using a modular architecture in which each functional component is implemented and tested independently before being integrated into the complete system. This modular approach improves maintainability, scalability, and reliability while ensuring that each feature functions correctly before deployment.

The overall system architecture follows the workflow:

User → Web Browser → Flask Web Application → Authentication Module → Functional Modules (Dashboard, AI Financial Advisor, Financial Chatbot, Fraud Detection, Document Management) → SQLite Database → Result Display → Activity Logs and Admin Dashboard

This architecture enables all project functionalities to operate seamlessly within a single integrated web application, providing users with a secure, intelligent, and efficient platform for personal financial management and digital banking assistance.

3.2 USER INTERFACE AND DASHBOARD

The User Interface and Dashboard of the Smart Financial Advisor and Secure Banking Assistant is developed as a web-based application using the Flask framework. The interface serves as the primary point of interaction between the users and the various functional modules of the system. The Flask application hosts the web pages and enables users to securely access financial services through a browser-based interface.

The main interface is designed with a modern and user-friendly layout that provides secure login and registration facilities. After successful authentication, users are redirected to a personalized dashboard where they can view their financial information, including monthly income, savings, expenses, and financial goals. The dashboard acts as the central navigation point of the application and allows users to access all available services from a single platform.

The application follows a structured project architecture in which the templates folder contains the HTML pages, the static folder stores CSS, JavaScript, and image files, and the backend logic is implemented in the Flask application. The SQLite database stores user information, financial records, uploaded documents, and activity logs.

The dashboard provides direct access to the major functional modules of the system, including the AI Financial Advisor, Financial Chatbot, Fraud Detection, and Document Management modules. Each module is accessible through dedicated navigation options, enabling users to switch seamlessly between services without leaving the application.

The AI Financial Advisor interface enables users to enter or update their financial details and receive personalized financial recommendations based on their income, savings, expenses, and financial goals. The Financial Chatbot provides an interactive platform where users can ask finance-related questions and receive instant responses. The Fraud Detection interface allows users to submit suspicious messages or transaction details for analysis, helping them identify potential online scams. The Document Management interface enables users to securely upload, view, and manage important financial documents.

For system administration, a dedicated Admin Dashboard is provided. This interface allows administrators to monitor registered users, view activity logs, manage user records, analyze financial statistics, generate charts, and export

reports in PDF and Excel formats. These administrative features simplify system monitoring and improve overall data management.

The user interface and dashboard are developed and tested as part of the Flask application before complete system integration. After deployment, users can access the application through any modern web browser, where the Flask server processes requests, communicates with the SQLite database, and displays the required information through an intuitive and responsive interface.

The integrated dashboard ensures that all financial management services are available through a single web application, providing users with a secure, efficient, and user-friendly platform for personal finance management and digital banking assistance.

3.3 USER AUTHENTICATION

The User Authentication module is responsible for providing secure access to the Smart Financial Advisor and Secure Banking Assistant. It verifies the identity of users before allowing them to access the application's financial services and ensures that only authorized users and administrators can use the system.

The authentication process is fully integrated into the Flask-based web application. Users access the system through the login page by entering their credentials, such as username and password. New users can create an account through the registration page by providing the required personal and financial information. The entered credentials are validated against the records stored in the SQLite database before access is granted.

After successful authentication, users are redirected to their personalized dashboard, where they can view and manage their financial information. From the dashboard, users can access the major functional modules of the application, including the AI Financial Advisor, Financial Chatbot, Fraud Detection, Document Management, and Profile Management. This integrated

approach allows users to access all available services through a single secure platform.

The application also provides a separate authentication mechanism for the Administrator. After successful admin login, the administrator is redirected to the Admin Dashboard, where they can manage registered users, monitor user activities, view financial statistics, generate charts, and export reports in PDF and Excel formats. This separation of user and administrator roles improves system security and ensures controlled access to administrative functions.

The authentication module is connected to the Activity Log feature, which records important user actions such as login, document uploads, financial advice requests, chatbot interactions, and fraud detection activities. These records are stored in the database and can be viewed through the administrative interface for monitoring and auditing purposes.

The authentication function therefore serves as the secure entry point to the application, ensuring that every user is properly verified before accessing financial services. By integrating secure login, role-based access control, and activity monitoring into a unified web application, the system provides a reliable and organized workflow from user authentication to financial service utilization while maintaining data security and system integrity.

3.4 KYC AND DIGITAL DOCUMENT PROCESSING

The Document Management and Digital Document Processing module is developed to enable users to securely upload, store, and manage important financial documents through the Smart Financial Advisor and Secure Banking Assistant. This module is fully integrated with the Flask-based web application, allowing document management to be performed within the same platform as the other financial services.

Users can access the Document Management section from the main dashboard and upload supported documents such as PDF, DOC, DOCX, JPG,

JPEG, and PNG files. The selected document is received by the Flask application and securely stored in the designated upload directory. The system maintains a record of uploaded documents, enabling users to access and manage their financial files whenever required.

The module is designed to organize uploaded documents efficiently while maintaining their association with the corresponding user account. Document upload activities are also recorded in the system's Activity Log, allowing administrators to monitor document-related actions and ensuring proper tracking of user interactions. This integration improves document organization and provides a secure method for storing important financial records.

The complete document management process follows the workflow:

User → Login → Dashboard → Document Management Module → Select Document → Upload Document → Secure Storage → Database Update → Activity Log → Document List Display

By integrating document management with the web-based application, the system provides a secure and centralized platform for uploading, storing, and managing financial documents. This approach eliminates the need for separate document management software, improves accessibility, and enhances the overall user experience while maintaining the security and integrity of uploaded files.

3.5 AI FINANCIAL ADVISOR

The AI Financial Advisor is one of the core modules of the Smart Financial Advisor and Secure Banking Assistant. It is developed to provide users with intelligent and personalized financial guidance through an interactive web-based interface. The module enables users to receive financial recommendations based on their financial information, helping them make informed decisions regarding savings, budgeting, investments, and future financial goals.

The AI Financial Advisor is integrated into the Flask-based web application and is accessible directly from the user dashboard. After logging

into the system, users can enter or update financial details such as monthly income, savings, expenses, and financial goals. The application processes this information and forwards the request to the integrated Groq API, which generates AI-powered financial recommendations. The generated response is then returned to the Flask application and displayed on the advisor page in an easy-to-understand format.

The module is designed to provide guidance on various financial topics, including budgeting, emergency fund planning, savings strategies, investment planning, debt management, retirement planning, and achieving personal financial goals. By analyzing the user's financial profile, the system delivers recommendations that are more relevant to the individual's financial situation.

The complete processing workflow follows the sequence:

User Login → Dashboard → AI Financial Advisor → Enter Financial Details/Query → Flask Application → Groq API Processing → AI Financial Recommendation → Display Result → Activity Log

The AI Financial Advisor is fully integrated with the overall application, allowing users to obtain financial guidance without accessing external services or separate applications. User interactions with the advisor are recorded in the Activity Log, enabling administrators to monitor system usage while maintaining a history of advisory requests.

By incorporating artificial intelligence into the financial management platform, the system provides users with quick, reliable, and personalized financial guidance through a single integrated web application. This enhances financial awareness, supports better financial decision-making, and improves the overall user experience.

3.6 FRAUD DETECTION

The Fraud Detection module is one of the key security features of the Smart Financial Advisor and Secure Banking Assistant. It is developed to help users identify potentially fraudulent financial messages, phishing attempts, fake

banking notifications, and suspicious transaction requests. The module is fully integrated with the Flask-based web application and can be accessed directly from the user dashboard.

Users can open the Fraud Detection page and paste suspicious content such as SMS messages, emails, UPI payment requests, website links, or other financial communications into the provided input field. The Flask application receives the submitted information and forwards it to the fraud detection module for analysis. The module examines the content for common fraud indicators, including requests for confidential information, suspicious links, urgency-based scams, fake rewards, and impersonation attempts.

After completing the analysis, the system classifies the submitted content according to its potential risk level and generates an appropriate warning or recommendation. The result is then displayed on the same web interface, allowing users to understand whether the message appears to be safe or potentially fraudulent.

The complete processing workflow follows the sequence:

User Login → Dashboard → Fraud Detection Module → Enter Suspicious Message/Transaction Details → Flask Application → Fraud Analysis → Risk Assessment → Result Display → Activity Log

The fraud detection activity is also recorded in the system's Activity Log, enabling administrators to monitor usage and maintain a history of fraud analysis requests. This allows previous fraud detection activities to be reviewed whenever required and supports effective system monitoring.

By integrating the Fraud Detection module into the common web application, the system provides users with a convenient and secure method for identifying potential financial scams without relying on separate security tools. This feature enhances user awareness, promotes safe digital banking practices, and contributes to protecting users from online financial fraud.

CHAPTER 4

HARDWARE AND SOFTWARE REQUIREMENTS

4.1 HARDWARE COMPONENTS

The hardware components of the **Smart Financial Services Kiosk with AI Advisory System** are selected to support the operation, user interaction, storage, connectivity, and physical integration of the complete kiosk prototype. The Raspberry Pi 4 Model B with 4 GB RAM serves as the main computing platform of the system. A 32 GB microSD card is used to store the Raspberry Pi operating system, application files, and other required software resources.

The kiosk uses a display interface through which the user accesses the web-based application and interacts with the available functional modules. The Raspberry Pi runs the Flask application and makes the website interface available for accessing the AI Financial Advisor, KYC and Document Upload, Fraud Detection, and User History modules.

A suitable power supply is provided for the Raspberry Pi and other required system components. Network connectivity is used for accessing the AI service and supporting communication between the Raspberry Pi and the web-based interface. The hardware components are assembled within the kiosk structure so that the computing unit, display arrangement, power connections, and required cables operate as a single integrated prototype.

Table 4.1 Hardware Components Used in the Project

S. No.	Component	Quantity	Purpose
1	Raspberry Pi 4 Model B – 4 GB RAM	1	Main computing platform
2	32 GB microSD Card	1	Operating system and project storage
3	Display Device	1	User interaction with the kiosk

			Interface
4	Raspberry Pi Power Adapter	1	Power supply for the Raspberry Pi
5	Network Connectivity Arrangement	1	Internet access and system communication
6	Connecting Cables	As required	Hardware and power connections
7	Kiosk Structure	1	Physical mounting and integration of the system

4.1.1 RASPBERRY PI 4 MODEL B (4 GB RAM)



Figure 4.1 Raspberry Pi 4 Model B

The Raspberry Pi 4 Model B with 4 GB RAM is used as the main computing platform of the Smart Financial Services Kiosk. It runs the Flask-based web application and supports the integration of the different functional modules. The Raspberry Pi processes user requests and manages the operation of the kiosk application through the web-based interface.

4.1.2 32 GB MICROSD CARD

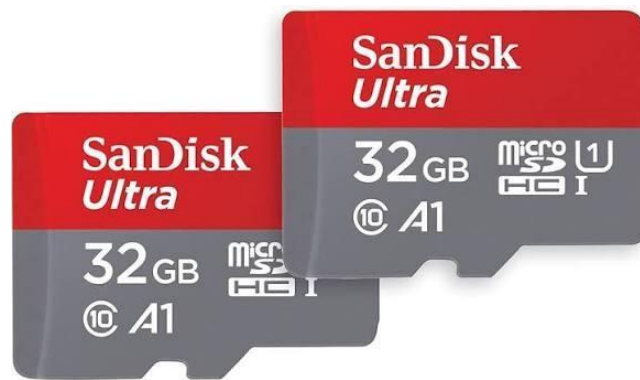


Figure 4.2 32 GB MicroSD Card

The 32 GB microSD card is used as the primary storage device for the Raspberry Pi. It stores the Raspberry Pi OS, project application files, and required software packages. The operating system is installed on the microSD card using Raspberry Pi Imager, and the card is inserted into the Raspberry Pi to boot and operate the complete kiosk application.

4.1.3 USB-C POWER ADAPTER



Figure 4.3 USB-C Power Adapter

The USB-C power adapter is used to provide the required electrical power to the Raspberry Pi 4 Model B. A 5 V, 3 A power adapter is used to support stable operation of the Raspberry Pi while running the kiosk application. It supplies continuous power to the main computing unit during system operation and testing.

4.1.4 SAMSUNG GALAXY TAB A11



Figure 4.4 Samsung Galaxy Tab A11

The Samsung Galaxy Tab A11 is used as the main user interaction interface of the Smart Financial Services Kiosk. Users access the web-based application through the tablet browser and interact with the available financial service modules using the touchscreen. The tablet allows users to enter details, submit financial queries, upload documents, and view the results generated by the system.

4.1.5 NETWORK CONNECTIVITY ARRANGEMENT

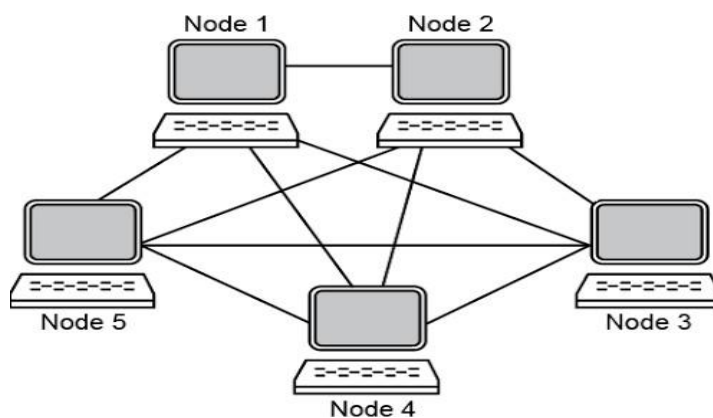


Figure 4.5 Network Connectivity Arrangement

The network connectivity arrangement is used to connect the Raspberry Pi and the Samsung Galaxy tablet to the same network. The Raspberry Pi runs the

Flask-based web application, while the tablet accesses the application through its web browser using the Raspberry Pi network address. Internet connectivity is also required for communication with the Groq API used by the AI Financial Advisor.

4.1.6 CONNECTING CABLES



Figure 4.6 Connecting Cables

The connecting cables are used to provide the required power and connections between the components of the Smart Financial Services Kiosk. They support the connection of the Raspberry Pi with the power source and other required system arrangements. The cables are organized inside the kiosk structure to ensure proper and continuous operation of the complete system.

4.1.7 KIOSK STRUCTURE



Figure 4.7 Kiosk Structure

The kiosk structure forms the main physical body of the Smart Financial Services Kiosk. It is designed to support and arrange the Raspberry Pi, Samsung Galaxy tablet, power connections, and other required components in an organized manner. The structure provides a suitable enclosure for integrating the hardware components and allows users to interact with the web-based application through the mounted tablet.

4.2 SOFTWARE COMPONENTS

4.2.1 RASPBERRY PI OS

The Raspberry Pi OS is used as the operating system for the Raspberry Pi 4 Model B. It provides the environment required to install the project dependencies, organize the application files, and run the Flask-based web application. The operating system is installed on the 32 GB microSD card and supports the execution of the Smart Financial Services Kiosk application and its integrated software modules.

4.2.2 PYTHON

Python is used as the main programming language for developing the backend of the Smart Financial Services Kiosk. It is used to run the Flask application, process user requests, connect the functional modules, and support communication with the Groq API. Python also provides the environment required to integrate the different software functions of the kiosk into a single web-based application.

4.2.3 FLASK

Flask is used as the web framework for developing the Smart Financial Services Kiosk application. It connects the website interface with the backend functions, manages navigation between the different modules, receives user inputs, and displays the corresponding results.

4.2.4 HTML

HTML is used to create the structure of the web pages in the kiosk application. It defines the content, buttons, input fields, forms, and other elements required for

users to interact with the different financial service modules.

4.2.5 CSS

CSS is used to control the appearance and layout of the web-based interface. It improves the presentation of the pages and organizes the visual elements of the kiosk application.

4.2.6 GROQ API

The Groq API is used to support the AI Financial Advisor module. It processes the financial queries entered by the user and returns the generated responses to the Flask application, which then displays them through the web-based interface.

4.2.7 GROQ PYTHON PACKAGE

The Groq Python package is used to connect the Python-based kiosk application with the Groq API. It enables the AI Financial Advisor module to send user queries to the AI service and receive the generated responses.

4.2.8 PYTHON-DOTENV

The python-dotenv package is used to manage the API configuration required by the application. It allows the Groq API key to be stored separately in an environment file and accessed by the Python application when required.

4.2.9 REQUESTS

The requests package supports communication between the application and external web-based services. It is used wherever the application requires HTTP-based data exchange during system operation.

4.2.10 PYTHON VIRTUAL ENVIRONMENT

A Python virtual environment is used to create an isolated environment for the Smart Financial Services Kiosk application. It allows the required Python packages and dependencies to be installed and managed separately without affecting the system-level Python environment.

CHAPTER 5

WORKING PRINCIPLE

5.1 USER LOGIN AND DASHBOARD ACCESS

The working process of the Smart Financial Advisor and Secure Banking Assistant begins when the user accesses the web-based application through a web browser. The user enters the required login credentials on the login page, and the Flask application verifies the entered information with the SQLite database. If the credentials are valid, the user is granted secure access to the system.

After successful authentication, the user is redirected to the Dashboard, which serves as the central navigation interface of the application. From the dashboard, the user can access the available modules, including the AI Financial Advisor, Financial Chatbot, Fraud Detection, Document Management, and Profile Management. The dashboard provides a unified interface for accessing all financial services without requiring separate applications.

The basic working flow is:

User Access → Login Page → User Authentication → Dashboard → Select Required Module → Access Service

This process provides a secure and organized entry point to the application while ensuring that only authenticated users can access financial services.

5.2 Document Management and Upload Process

The Document Management module enables users to securely upload and manage important financial documents through the web application. From the dashboard, users can access the document management page and select supported files such as PDF, DOC, DOCX, JPG, JPEG, and PNG for upload.

The uploaded file is received by the Flask application, securely stored in the upload directory, and its details are recorded in the database. The upload activity is also maintained in the system's activity log for future reference.

The basic working flow is:

Dashboard → Document Management → Select Document → Upload Document → Secure Storage → Database Update → Activity Log → Document List Display

This process allows users to organize and manage financial documents securely within the same application.

5.3 AI FINANCIAL ADVISORY PROCESS

The AI Financial Advisor module provides personalized financial guidance based on the user's financial information. Users access the module from the dashboard and enter or update details such as income, savings, expenses, and financial goals.

The Flask application processes the submitted information and forwards it to the integrated Groq API, which generates AI-powered financial recommendations. The generated advice is then displayed on the advisor page.

The basic working flow is:

Dashboard → AI Financial Advisor → Enter Financial Details/Query → Flask Application → Groq API Processing → AI Recommendation → Display Result

This process enables users to receive intelligent financial guidance directly through the application, helping them make informed financial decisions.

5.4 Financial Chat Box

The Financial Chatbot module allows users to ask finance-related questions through an interactive interface. Users can enter questions related to budgeting, savings, investments, loans, insurance, or other financial topics.

The Flask application receives the query and sends it to the AI processing service, which generates an appropriate response. The chatbot's reply is then displayed instantly within the application.

The basic working flow is:

Dashboard → Financial Chatbot → Enter Question → Flask Application → AI Processing → Chatbot Response → Display Result

This module provides users with quick and interactive financial assistance whenever needed.

5.5 Fraud Detection Process

The Fraud Detection module helps users identify suspicious financial messages and potential online scams. Users access the module from the dashboard and submit suspicious SMS messages, emails, website links, or UPI payment requests.

The Flask application forwards the submitted content to the fraud detection module, where it is analyzed for phishing indicators, fake payment requests, suspicious links, and other fraud patterns. Based on the analysis, the system generates a risk assessment and displays an appropriate warning or recommendation.

The basic working flow is:

Dashboard → Fraud Detection → Enter Suspicious Content → Fraud Analysis → Risk Assessment → Result Display

This process assists users in recognizing fraudulent activities and promotes safer digital banking practices.

CHAPTER 6

COST ESTIMATION

S. No.	Component	Quantity	Unit Cost (₹)	Total Cost (₹)
1	Raspberry Pi 4 Model B – 4 GB RAM	1	12,200	12,200
2	32 GB microSD Card	1	820	820
3	Samsung Galaxy Tablet	1	20,000	20,000
4	Raspberry Pi Power Adapter	1	600	600
5	Network Connectivity Arrangement (USB Hub)	1	230	230
6	Connecting Cables	As required	300	300
7	Kiosk Structure (<i>Plywood, Colour Paper, Bolt & Nut, Double Side Tape, Fevicol</i>)	1	555	555
8	Cooling Fan	1	65	65
9	Buck Converter	1	75	75
10	Adapter	1	600	600
	TOTAL COST			₹35,445

Table 6.1 Cost Estimation

CHAPTER 7

RESULT DOCUMENTATION AND SYSTEM OUTPUT

The developed Smart Financial Advisor and Secure Banking Assistant integrates multiple financial management and security modules into a single web-based application. The system is developed using Python, Flask, SQLite, HTML, CSS, and JavaScript, providing users with a secure and user-friendly platform for managing personal finances. Users access the application through a web browser, where the Flask server processes requests and communicates with the SQLite database to provide the required services.

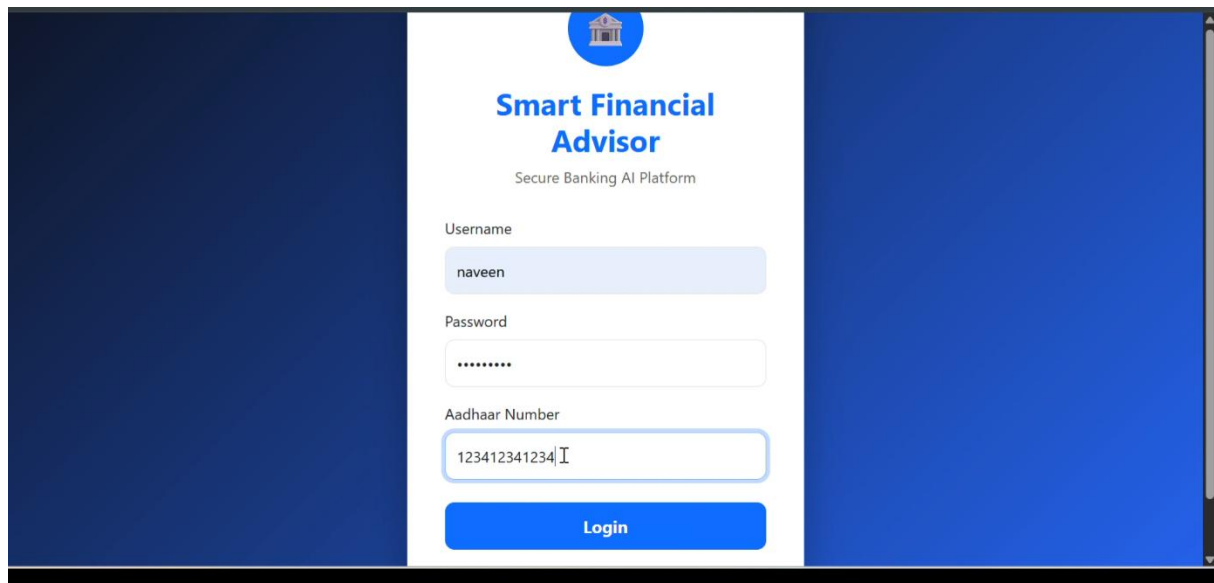
The system output is documented by testing each functional module individually and verifying its corresponding interface, processing, and generated results. The outputs demonstrate the successful implementation of the major features, including user authentication, dashboard navigation, AI Financial Advisor, Financial Chatbot, Fraud Detection, Document Management, and the Admin Dashboard.

7.1 LOGIN AND DASHBOARD OUTPUT

The Login interface serves as the secure entry point to the Smart Financial Advisor and Secure Banking Assistant. Users enter their registered login credentials through the authentication page, and the Flask application validates the information using the SQLite database. Upon successful authentication, users are granted secure access to the application.

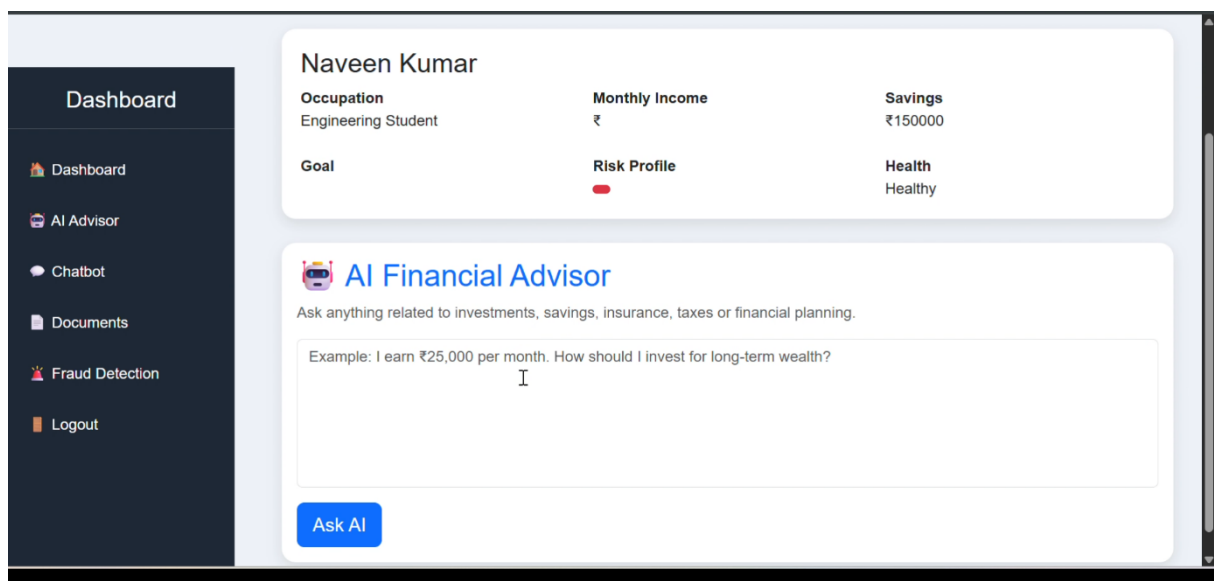
After logging in, the user is redirected to the Dashboard, which functions as the central navigation interface of the system. The dashboard displays the user's financial information and provides direct access to the application's major functional modules, including the AI Financial Advisor, Financial Chatbot, Fraud Detection, Document Management, and Profile Management. The dashboard also provides quick navigation between modules, allowing users to manage all financial services through a single integrated web application.

The successful display of the dashboard confirms that the authentication process, database connectivity, and navigation functions are operating correctly, providing users with a secure and organized environment for accessing the financial management services.



The login page features a central white card on a dark blue background. At the top of the card is a blue circular icon with a white building. Below the icon, the text 'Smart Financial Advisor' is displayed in bold blue font, followed by 'Secure Banking AI Platform' in a smaller grey font. The login form includes three input fields: 'Username' with the value 'naveen', 'Password' with masked characters '*****', and 'Aadhaar Number' with the value '123412341234'. A blue 'Login' button is positioned at the bottom of the form.

Figure 7.1 Login Page of the Smart Financial Services Kiosk



The main dashboard has a dark blue sidebar on the left with a 'Dashboard' header and a list of menu items: 'Dashboard', 'AI Advisor', 'Chatbot', 'Documents', 'Fraud Detection', and 'Logout'. The main content area is white and displays user information for 'Naveen Kumar'. This information is organized into two rows of three items each: 'Occupation' (Engineering Student), 'Monthly Income' (₹), 'Savings' (₹150000) in the first row, and 'Goal', 'Risk Profile' (indicated by a red bar), 'Health' (Healthy) in the second row. Below this is a section for the 'AI Financial Advisor', which includes a chatbot icon, the title 'AI Financial Advisor', and a prompt: 'Ask anything related to investments, savings, insurance, taxes or financial planning.' An example question is provided: 'Example: I earn ₹25,000 per month. How should I invest for long-term wealth?'. A text input field contains this example text, and a blue 'Ask AI' button is at the bottom.

Figure 7.2 Main Dashboard of the Smart Financial Services Kiosk

The obtained output confirms that the user can access the system through the login interface and navigate to the available functional modules from the main dashboard.

7.2 KYC AND DOCUMENT UPLOAD OUTPUT

The Document Management module enables users to securely upload and manage important financial documents through the web-based application. During system testing, the user accesses the Document Management module from the dashboard and uploads a supported document, such as a PDF, DOC, DOCX, JPG, JPEG, or PNG file, using the upload option provided on the interface.

After the document is selected, the Flask application receives the uploaded file, validates it, and stores it securely in the designated upload directory. The document information is then recorded in the SQLite database, and the upload activity is maintained in the system's activity log. Once the upload process is completed successfully, the application displays a confirmation message and updates the list of uploaded documents on the interface.

The successful output of this module confirms that the document upload, secure storage, database integration, and activity logging functions are operating correctly. This enables users to organize, store, and access their important financial documents through a single, secure, and integrated web application.

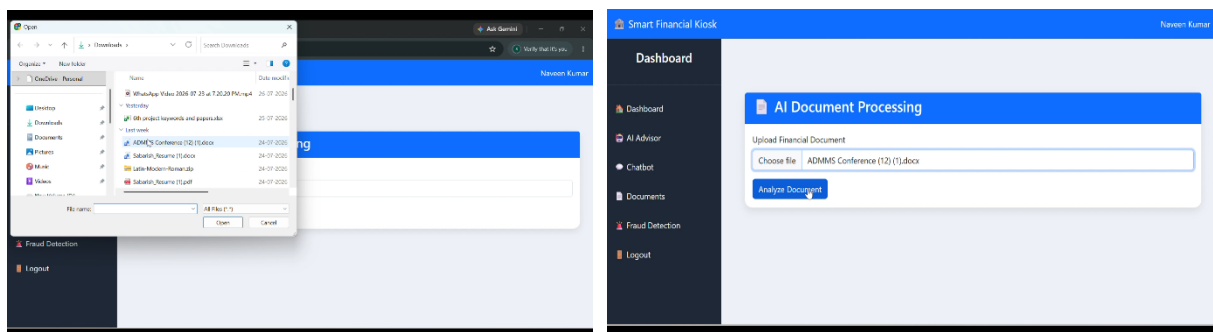


Figure 7.3 KYC and Document Upload Page and Document Upload

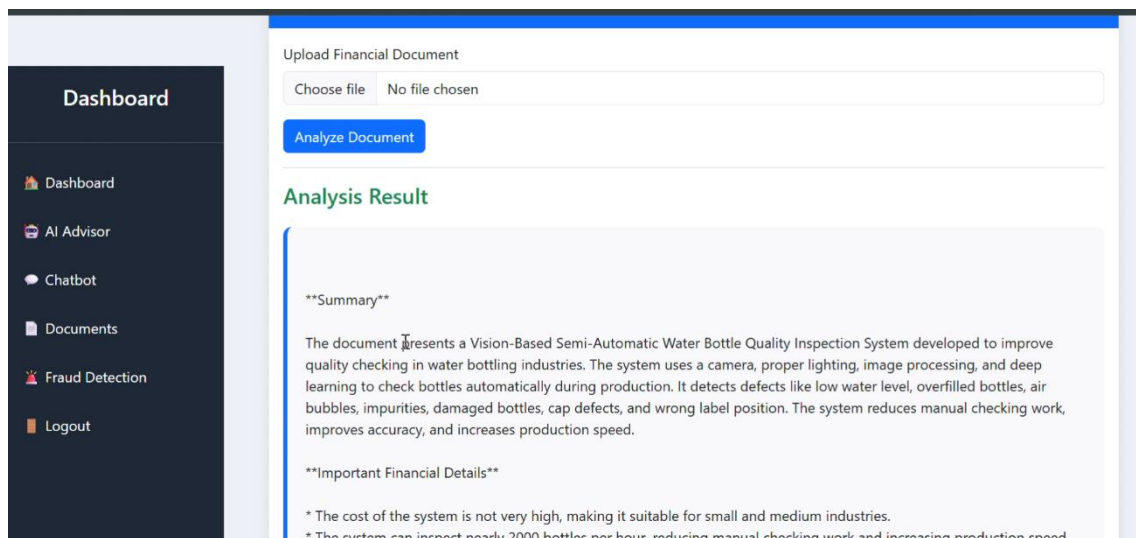


Figure 7.4 Result of Uploaded Document

The obtained output demonstrates the document submission process through the kiosk interface. The uploaded document is received by the application, and the corresponding result is displayed to the user through the web-based system.

7.3 AI FINANCIAL ADVISOR OUTPUT

The AI Financial Advisor module allows users to receive personalized financial guidance through the web-based application. During system testing, the user accesses the AI Financial Advisor from the dashboard and enters financial information or a finance-related query. The Flask application processes the request and forwards it to the integrated Groq API, which generates an AI-based financial recommendation. The generated response is then displayed on the same interface, providing users with relevant suggestions related to budgeting, savings, investments, and financial planning.

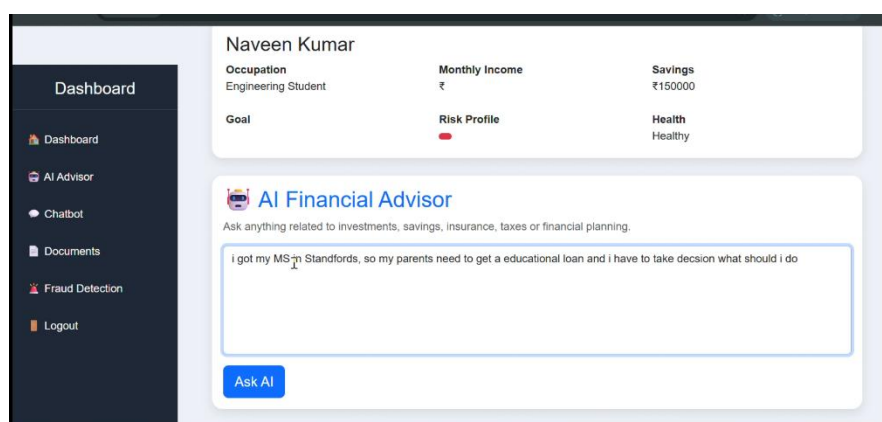


Figure 7.5 AI Financial Advisor Interface

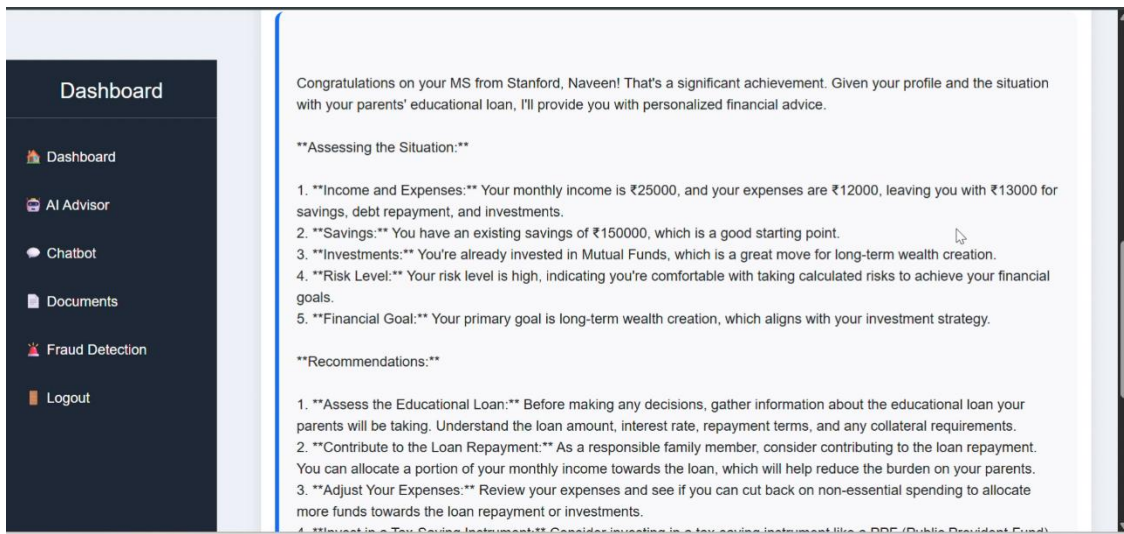


Figure 7.6 AI Financial Advisor Response

The obtained output demonstrates that the user can enter a financial query and receive the corresponding AI-generated guidance through the Smart Financial Services Kiosk interface.

7.4 FRAUD DETECTION OUTPUT

The Fraud Detection module allows the user to provide the required information through the web-based interface for fraud risk evaluation. During testing, the entered details are processed by the fraud detection module, and the corresponding risk result is displayed to the user.

Dashboard

Dashboard

AI Advisor

Chatbot

Documents

Fraud Detection

Logout

Fraud Detection

Account Holder: Naveen Kumar

Date: 26-07-2026

Time: 09:29 PM

100000

UPI

Salem

mobile

salem

sabarish

Check Fraud

Figure 7.7 Fraud Detection Interface

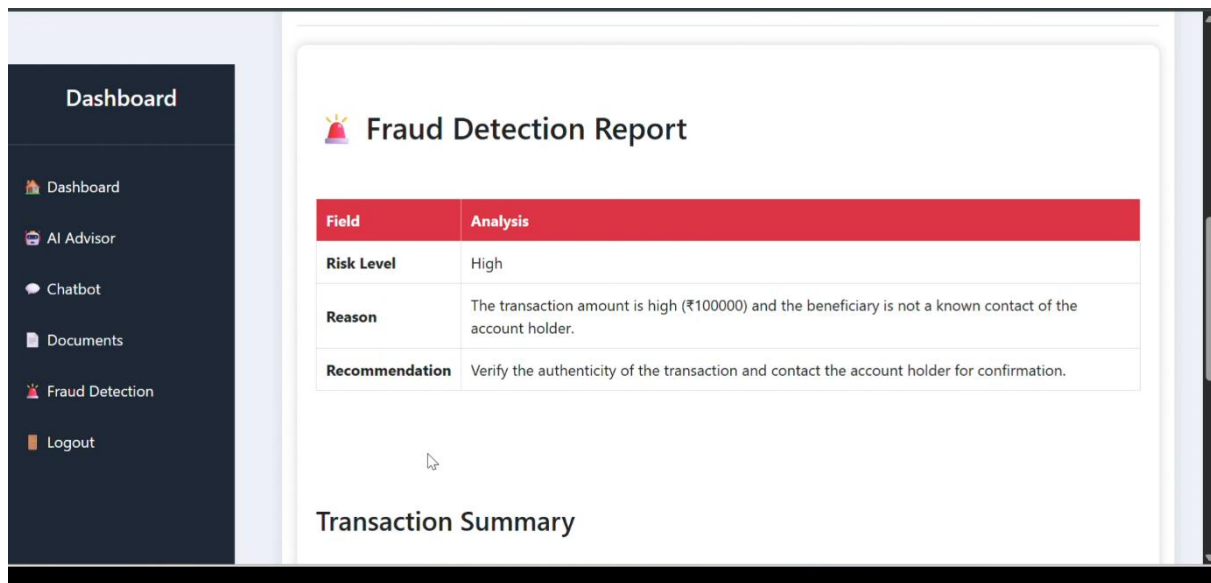


Figure 7.8 Fraud Detection Result

The obtained output demonstrates that the system can receive the required user information, perform the fraud risk evaluation, and display the corresponding result through the Smart Financial Services Kiosk interface.

CHAPTER 8

SCOPE OF FUTURE WORK

The Smart Financial Advisor and Secure Banking Assistant can be further enhanced by improving its existing modules and extending its capabilities to provide more intelligent, secure, and personalized financial management services. The current system integrates user authentication, financial management, AI-based financial advisory, financial chatbot, fraud detection, document management, and administrative monitoring into a single web-based application.

Future development can focus on enhancing the AI Financial Advisor by incorporating advanced machine learning models to provide more personalized financial recommendations based on user spending patterns, savings behavior, and financial goals. The Financial Chatbot can be upgraded to support multilingual communication, voice-based interaction, and more advanced conversational capabilities to improve user experience.

The Document Management module can be enhanced by integrating Optical Character Recognition (OCR) technology for automatic extraction of information from uploaded financial documents. Additional document verification mechanisms and cloud storage support can also be implemented to improve accessibility and security.

The Fraud Detection module can be further improved by incorporating machine learning techniques to detect phishing attacks, fraudulent messages, and suspicious financial transactions with greater accuracy. Real-time fraud monitoring and automatic security alerts can further strengthen the application's security features.

Future versions of the system can also include stronger authentication mechanisms such as Two-Factor Authentication (2FA), biometric authentication, encrypted data storage, and cloud database integration.

Additional features such as investment portfolio management, expense prediction, bill payment reminders, financial goal tracking, and integration with banking APIs can transform the application into a more comprehensive personal finance management platform.

These enhancements will improve the intelligence, security, scalability, and usability of the system while supporting a wider range of financial management applications.

8.1 ADVANTAGES

- Integrates multiple financial management services into a single web-based application.
- Provides secure user registration and authentication.
- Offers AI-based financial advice to support informed financial decision-making.
- Includes an interactive financial chatbot for answering finance-related queries.
- Helps users identify suspicious financial messages through the fraud detection module.
- Enables secure upload and management of important financial documents.
- Provides an intuitive dashboard for monitoring income, savings, expenses, and financial goals.
- Includes an administrator dashboard for user management, activity monitoring, analytics, and report generation.
- Supports report generation and data export in **PDF** and **Excel** formats.
- Provides a user-friendly, secure, and centralized platform for personal financial management.

8.2 APPLICATIONS

- Personal financial management systems.
- Digital banking and financial assistance platforms.
- Financial advisory and budgeting applications.
- Online fraud awareness and phishing detection systems.
- Secure financial document management solutions.
- Banking customer support and self-service portals.
- Educational institutions for demonstrating AI-based financial management systems.
- Financial literacy and awareness programs.
- Small business financial management applications.
- Research and development projects related to intelligent financial advisory systems.

CHAPTER 9

APPENDICES

The appendices provide supporting information related to the hardware, software, system flow, and user interface of the Smart Financial Services Kiosk with AI Advisory System. These details are included separately to support the technical documentation of the developed prototype.

APPENDIX 1

COMPONENTS & PROTOTYPE SPECIFICATION

1. Raspberry Pi 4 Model B Specifications

- Model: Raspberry Pi 4 Model B
- RAM: 4 GB
- Processor: Broadcom BCM2711, Quad-Core Cortex-A72
- Processor Speed: 1.5 GHz
- USB Ports: 2 × USB 3.0 and 2 × USB 2.0
- Network Connectivity: Wi-Fi, Bluetooth and Gigabit Ethernet
- Display Output: 2 × Micro-HDMI Ports
- Power Input: USB-C
- Storage Support: microSD Card
- Operating System: Raspberry Pi OS
- Application: Main computing platform of the kiosk

2. 32 GB microSD Card Specifications

- Storage Capacity: 32 GB
- Card Type: microSD
- Speed Class: Class 10
- Application: Raspberry Pi OS and project file storage
- Function: Boot and storage device for the Raspberry Pi

3. Raspberry Pi Power Adapter Specifications

- Output Voltage: 5 V DC
- Output Current: 3 A
- Connector Type: USB-C
- Input Supply: AC mains
- Application: Power supply for the Raspberry Pi 4 Model B

4. Samsung Galaxy Tablet Specifications

- Device Type: Android Tablet
- Operating System: Android
- Display Type: Touchscreen
- Connectivity: Wi-Fi
- User Input: Touch-based interaction
- Application: User interface for accessing the kiosk web application

5. Network Connectivity Arrangement Specifications

- Connection Type: Wireless Network
- Technology: Wi-Fi
- Connected Devices: Raspberry Pi and Samsung Galaxy Tablet
- Internet Requirement: Required for Groq API communication
- Local Network Function: Access to the Flask application through the Raspberry Pi network address
- Application: Communication between the kiosk interface and main computing platform

6. Connecting Cables Specifications

- Cable Type: Power and connection cables
- Connection Requirement: Based on the system setup
- Function: Power supply and required hardware connections

- Application: Integration of the kiosk hardware arrangement

7. Kiosk Structure Specifications

- Structure Type: Enclosed kiosk body
- Display Arrangement: Front-mounted tablet interface
- Internal Arrangement: Space for Raspberry Pi and power connections
- Cable Arrangement: Internal cable routing
- Application: Physical integration and mounting of the complete kiosk system

PROTOTYPE SPECIFICATION

- Prototype Name: Smart Financial Services Kiosk with AI Advisory System
- Main Computing Platform: Raspberry Pi 4 Model B with 4 GB RAM
- Storage: 32 GB microSD Card
- User Interface: Samsung Galaxy Android Tablet
- Application Type: Web-Based Application
- Web Framework: Flask
- Main Programming Language: Python
- AI Service: Groq API
- Main Functional Modules: AI Financial Advisor, KYC and Document Upload, Fraud Detection, and User History
- Network Requirement: Wi-Fi and Internet Connectivity

APPENDIX 2

SYSTEM FLOWCHART

The system flowchart represents the overall working sequence of the Smart Financial Services Kiosk. It shows the process from user access and dashboard navigation to the selection and operation of the available functional modules.

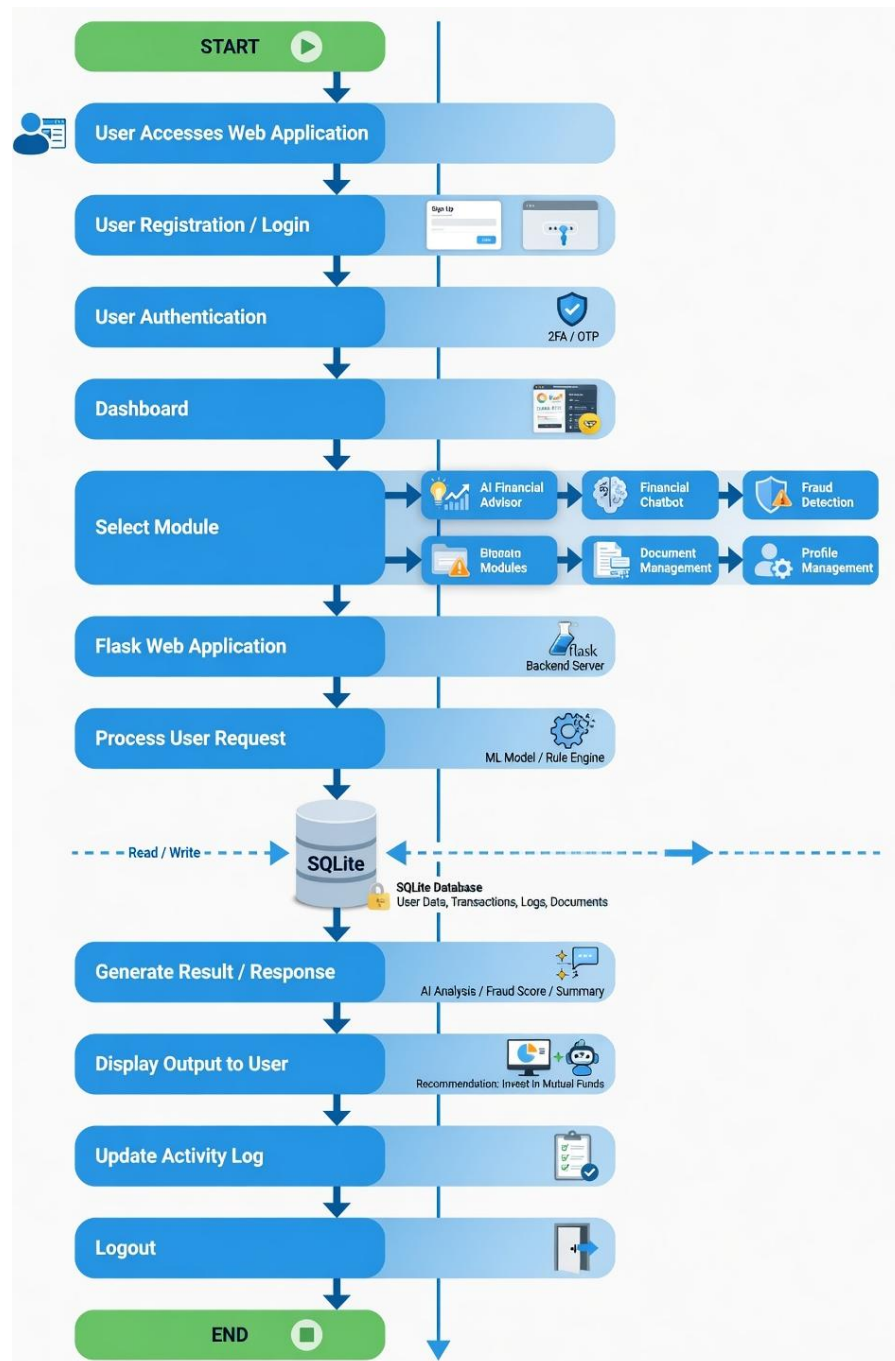


Figure A.1 Overall System Flowchart

APPENDIX 3

PROTOTYPE VIEWS



Figure A.2 Front View of the Smart Financial Services Kiosk



Figure A. 3 Side View of the Smart Financial Services Kiosk



Figure A.4 Top View of the Smart Financial Services Kiosk



Figure A.5 Back View of the Smart Financial Services Kiosk

APPENDIX 4

PSEUDO CODE

Algorithm 1: System Initialization

```
BEGIN
Initialize Flask Application
Load Configuration Files
Connect to SQLite Database
IF Database Connection = SUCCESS THEN
    Load AI Advisor Module
    Load Fraud Detection Module
    Load Document Management Module
    Display Login Page
ELSE
    Display "System Initialization Failed"
    EXIT
END IF
```

Algorithm 2: User Authentication

```
BEGIN
Read Username and Password
IF Username = NULL OR Password = NULL THEN
    Display "Enter Valid Credentials"
ELSE
    Search User in Database
    IF User Exists THEN
        Create User Session
        Redirect to Dashboard
    ELSE
        Display "Invalid Login"
```

```
END IF  
END IF  
END
```

Algorithm 3: Document Management

```
BEGIN  
Receive Uploaded Document  
IF File Type is Valid THEN  
    Save File to Upload Folder  
    Store File Details in Database  
    Update User Activity  
    Display "Upload Successful"  
ELSE  
    Display "Invalid File Format"  
END IF  
END
```

Algorithm 4: AI Financial Advisor

```
BEGIN  
Receive Financial Query  
IF Query is Empty THEN  
    Display "Enter Financial Query"  
ELSE  
    Send Query to Groq API  
    Receive AI Response  
    Display Recommendation  
    Store Chat History  
END IF  
END
```

Algorithm 5: Fraud Detection

```
BEGIN
Receive Financial Message
Preprocess Input Message
Analyze Message using Fraud Detection Rules
IF Message is Fraudulent THEN
    Display "Fraud Detected"
ELSE
    Display "Message is Safe"
END IF
Store Result in Database
END
```

CHAPTER 10

CONCLUSION

The Smart Financial Advisor and Secure Banking Assistant was successfully developed as a secure and intelligent web-based application that integrates multiple financial management services into a single platform, combining user authentication, financial management, AI-based financial advisory, a financial chatbot, fraud detection, document management, and administrative monitoring through a unified and user-friendly interface. The application is developed using Python, Flask, SQLite, HTML, CSS, and JavaScript, providing a reliable and efficient platform for managing personal financial information, allowing users to securely access the system, monitor their income, savings, expenses, and financial goals, receive personalized financial recommendations through the Groq AI API, detect potentially fraudulent financial messages, and securely upload important financial documents. The integrated Admin Dashboard further enables efficient user management, activity monitoring, analytics, and report generation. The developed system demonstrates the successful integration of artificial intelligence, secure data management, fraud awareness, and digital financial services within a single web application, providing a practical solution for improving financial literacy, supporting informed financial decision-making, and enhancing digital banking security. Overall, the project achieves its objectives by delivering a secure, user-friendly, and scalable financial management platform that can be further enhanced with advanced AI capabilities, cloud integration, and additional financial services in future versions.

CHAPTER 11

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